Module 4 : Smart Technologies

102030622 – Smart Cities Planning and Management

Transportation System Management in Smart Cities

What is Smart Transportation?

According to the US Department of Transportation, “Intelligent Transportation Systems (ITS) apply a variety of technologies to monitor, evaluate, and manage transportation systems to enhance efficiency and safety.”

Main Concepts : management, efficiency, and safety

Smart transportation uses new and emerging technologies to make moving around a city more convenient, more cost effective (for both the city and the individual), and safer.

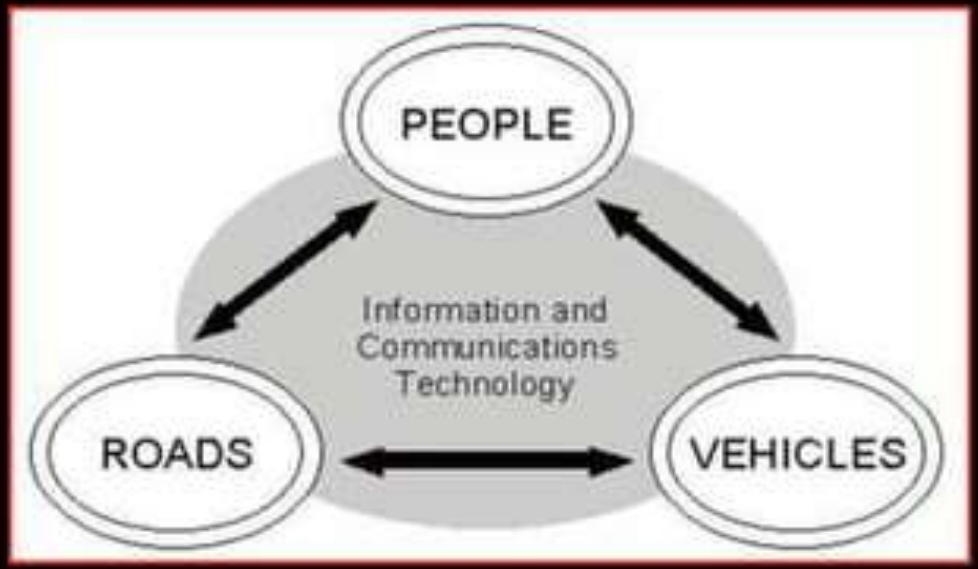
What is Smart Transportation?

Intelligent transportation system is the application of sensing, analysis, control and communications technologies to ground transportation in order to improve safety, mobility and efficiency of road transportation systems.

Main goal is to reducing congestion on city streets.

With sensors, advanced communication technologies, automation & high-speed networks.

ITS is an emerging transportation system which is comprised of an advanced information and telecommunications network for users, roads and vehicles.



ITS is the integrated application of advanced Technologies using electronics, computers, communications, and advanced sensors. These applications provide travelers with important information while improving the safety and efficiency of the transportation system.

GOALS OF ITS

- Improved Safety
- Increased and Higher Quality Mobility
- Reduced Environmental Impact
- Improved Energy Efficiency
- Improved Economic Productivity
- Make transportation systems more efficient, secure, and safer through the use of information, communications and control technologies.
- Improve the attractiveness of public transport.
- Tackle rising congestion which increases travel times and industry costs.

BENEFITS OF ITS

a) For Government

1. Instant tracking of any vehicle.

2.Identification and location of vehicles involved in crimes, tax default vehicles, accident involved vehicles and stolen vehicles.

3.Message to any number of vehicles in almost no time.

4.Easy traffic diversion.

5.Effective enforcement with less manpower and cheaper enforcement cost.

6.Advance traveler information system.

7.100% accurate toll collection.

b) For Users -

1. Easy instant toll payment.

2.Safety requirements and anti-collision devices integration.

3.Fleet management at low cost.

4.Increase in fuel efficiency.

5.Advance traveler information/traffic congestion information and dynamic navigation of vehicles.

6.Anti-theft systems.

COMPONENTS

Highway Traffic Management System (HTMS)

Weight-in-motion (WIM)

Toll Collection Equipment (TCE)

Highway Traffic Management System (HTMS)

Highway Traffic Management System (HTMS) is also known as Advance Traffic Management system (ATMS).

It consists of various traffic and incident monitoring devices installed along the highway and connected to a central control room via a communication backbone.

Once a incident is detected or distress call is received, the control room sends appropriate help (ambulance or crane).

The various components of HTMS are:

Emergency Call Boxes (ECB)

Closed Circuit Television (CCTV) camera

Automatic Traffic Counter cum Classifier (ATCC)

Weather Station

Variable Message Sign (VMS)

Mobile radio

The Optic Fibre Cable (OFC) network

ECB are installed on the highway- one pair at every two kilometres

(one ECB on either side of carriageway).

The ECB consists of a hands free speakers and microphones and audio processing, transmission and signalling circuitry.

Tools used as Emergency Call Boxes (ECB)

Closed Circuit Television (CCTV) camera

CCTV cameras are installed along the highway at incident hotspots.

CCTV cameras usually cover 1-2 km

Tools used as Closed Circuit Television (CCTV) camera

Automatic Traffic Counter cum Classifier (ATCC)

ATCC is a device which is used to measure the volume and class (car / truck / LCV / multi-axle) of vehicular traffic on the road.

ATCC is used to gather traffic data, monitor volumes of toll plaza traffic.

Tools used as Automatic Traffic Counter cum Classifier (ATCC)

Weather Station

A weather station is usually installed at a toll plaza to record characteristics like humidity, rain, temperature, road temperature, visibility, wind speed and direction etc.

Helpful to displayed to road user through Variable Message Sign (VMS) for information and decision making

Tools used as Weather Station

Variable Message Sign (VMS)

It is a screen mounted on a gantry.

The screen display is typically made of LEDs (light emitting diodes).

The message displayed on VMS can be changed by the Call Center Representative (CCR) operator.

Main goal is to provide the road user about the information on the highway like road closure, accidents ahead occupancy, weather information etc.

Tools used a Variable Message Sign (VMS)

Mobile radio

The mobile radio network consists of hand-held walkie-talkies and vehicle mounted units which enable the emergency personnel to communicate with the CCR as well as with other team members on the field.

WEIGHT-IN-MOTION (WIM)

Weight-in-motion (WIM) is a system which can measure the weight of a moving vehicle with great accuracy.

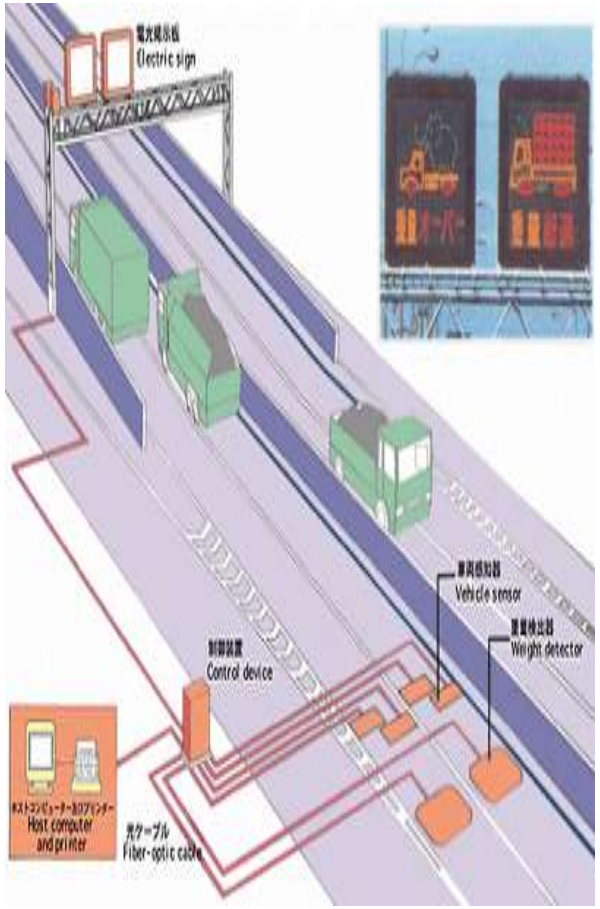
It consists of In-road sensors connected to a roadside unit.

As a moving truck goes over the sensors, its axle pressurizers the sensor which induces a change in sensor parameter (like charge or strain) which is then measured by the roadside electronic unit to give axle weight.

Axle weights are then added up to give GVW (gross vehicle weight).

Types: piezoelectric sensors, bending plates, hydraulic pressure sensors, capacitive sensors, single load cells etc.,

Weight-in-motion (WIM) Techniques



Toll Collection Equipment (TCE)

TCE is installed at toll plazas to assist in toll fee collection.

It consists of the following components:

Toll lane equipment consisting of receipt printer, booth computer, bar code reader, smart card reader, tag reader / transponder, traffic light, barrier gate & lane open / close signals.

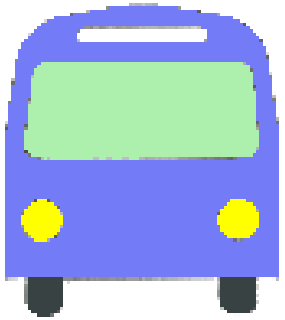
Automatic Vehicle Classification (AVC) system consisting of in-road sensors which automatically classify the vehicle as it passes over it.

Toll Collection Techniques

INTELLIGENT INFRASTRUCTURE

Arterial Management

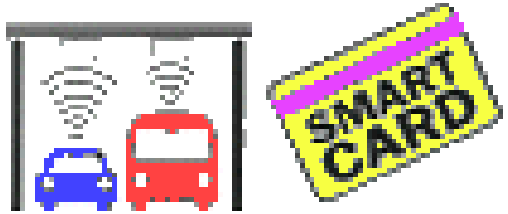
Freeway Management



Transit Management

Incident management

Emergency management



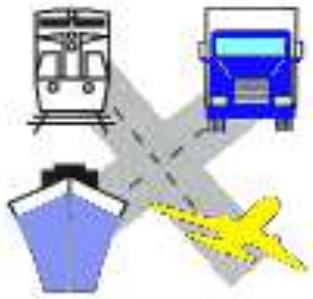
Electronic payment and pricing

INTELLIGENT INFRASTRUCTURE

Traveler information

Information Management

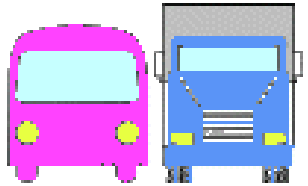
Crash Prevention and Safety



Inter-Modal Freight

Roadway Operation and Maintenance

Road Weather Management



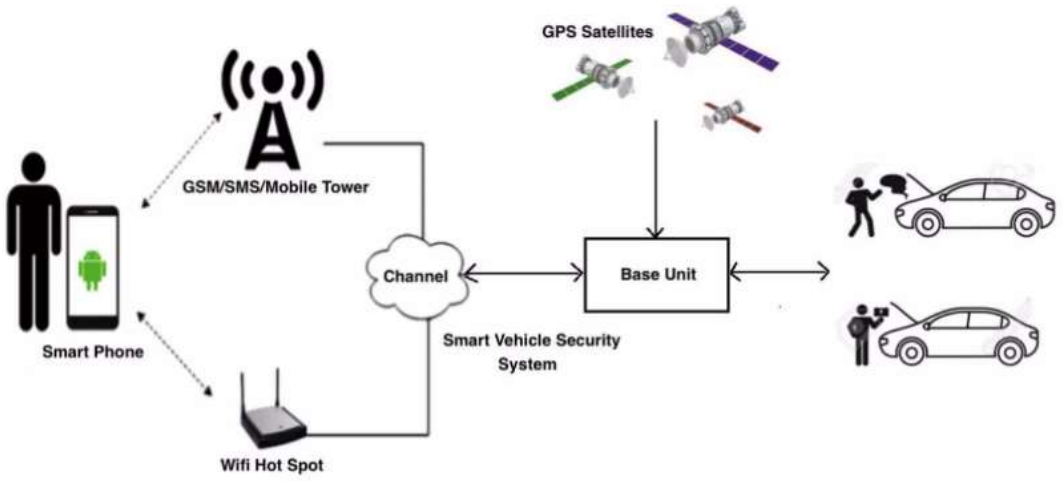
Commercial Vehicle operation

‘Smart transportation’ refers to the integrated application of modern technology & management strategies into transportation systems.

1. Car navigation
2. Traffic signal control systems
3. Automatic number plate recognition
4. Speed cameras

Key features of smart system

- Smart Vehicles
- Sensors
- Navigation system
- Chip
- Auto Breaking System



Smart vehicle is connected with the server and all information of the vehicle is stored on server.

User can be locate of the vehicle from any place by login of server and find current position.

https://www.digi.com/blog/post/introduction-to-smart-transportation-benefits#:~:text=Smart%20Transportation%20provides%20rapid%20insights,communicate%20more%20effectively%20with%20other

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